

Key

Physics 214/224, Quiz 4

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04/12/2019

For grading purposes

Problem 1	
Problem 2	
Problem 3	
Problem 4	
Studio	
Total:	

WID:_____

NAME:_____

Instructions. Print and sign your name on this quiz and on your scantron card. In doing so, you are acknowledging the KSU Honor Code: "On my honor as a student I have neither given nor received unauthorized aid on this academic work."

For two quick points, circle your studio instructor:

Smith (TU 7:30) Lohman (TU 9:30) Maravin (TU 11:30)

Maravin (TU 1:30) Smith (TU 3:30)

Weliweriya (WF 9:30) Weliweriya (WF 11:30) Smith (WF 1:30)

Work alone and answer all questions. Part I questions must be answered on the Scantron cards. Put your name on your card. Color in the correct box completely for every answer with a pencil. If you make a mistake, erase thoroughly. **Don't forget to color in the boxes for your WID. If we have to correct this by hand we may take off five points from your score!** Part II must be answered in the space provided on the test. The last page is a detachable equation sheet. You may use a calculator. You may ask the proctors questions of clarification. Show your work in a clear and organized manner. You may receive partial credit for partial solutions. Solutions lacking supporting calculations will receive no points.

BRACE YOURSELVES

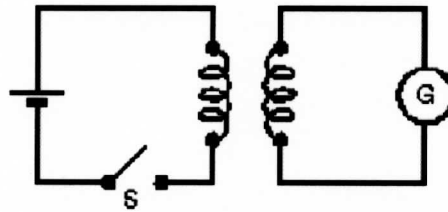


Part I: All questions to be answered on your Scantron card (48 points total)

1. (3 points) Faraday's law states that an induced emf is proportional to

- (a) the rate of change of the magnetic field
- (b) the rate of change of the electric field
- ☒ (c) the rate of change of the magnetic flux
- (d) the rate of change of the electric flux
- (e) zero

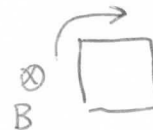
2. (3 points) In the circuit shown below, there will be a non-zero reading in galvanometer G:



- (a) only just after S is closed
- (b) only just after S is opened
- (c) only while S is kept closed
- (d) never
- ☒ (e) only just after S is opened or closed

3. (3 points) A square loop of wire lies in the plane of the page. A decreasing magnetic field is directed into the page. The induced current in the loop is:

- (a) counterclockwise
- ☒ (b) clockwise
- (c) zero
- (d) up the left edge and from right to left along the top edge
- (e) through the middle of the page

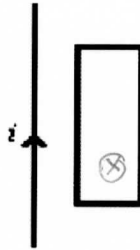


4. (3 points) The stored energy in an inductor

- (a) depends, in sign, upon the direction of the current
- (b) depends on the rate of change of current
- (c) is proportional to the square of inductance
- (d) has units J/H
- ☒ (e) is none of the above

$$PE = L \frac{I^2}{2}$$

The next two questions refer to the image below that depicts a long straight wire in the plane of a rectangular conducting loop.



5. (3 points) The straight wire carries a constant current i , as shown. While the wire is being moved toward the loop, the current in the loop is:
- (a) zero
 (b) clockwise
 (c) counterclockwise
 (d) clockwise in the left side and counterclockwise in the right side
 (e) counterclockwise in the left side and clockwise in the right side
6. (3 points) In this case, the straight wire carries an increasing current in the direction shown. The current in the loop is
- (a) zero
 (b) clockwise
 (c) counterclockwise
 (d) clockwise in the left side and counterclockwise in the right side
 (e) counterclockwise in the left side and clockwise in the right side
7. (3 points) A coil of wire, having 5 turns, has an inductance L . The inductance of a similar coil having 20 turns is:
- (a) $4L$
 (b) $L/4$
 (c) $16L$
 (d) $L/16$
 (e) L
8. (3 points) A 3.5 mH inductor and a 4.5 mH inductor are connected in parallel. The equivalent inductance is
- (a) 0.13 mH
 (b) 0.51 mH
 (c) 1.0 mH
 (d) 2.0 mH
 (e) 8.0 mH

$B \uparrow$

$$L \sim N^2$$

$$\frac{1}{\frac{1}{4.5} + \frac{1}{3.5}} = 1.97 \text{ mH} \approx 2 \text{ mH}$$

9. (3 points) The image below depicts a self-induced emf direction (to the right) and the direction of the current i at some point of time. The current i is



- (a) increasing
 (b) decreasing *Since EMF is in the direction of the current (helps with i)*
 (c) not changing
 (d) either increasing or decreasing
 (e) one of the three: increasing or decreasing or not changing
10. (3 points) Which of the following has the greatest effect in decreasing the oscillation frequency of an LC circuit? Using instead
- (a) $L/2$ and $C/2$
 (b) $L/2$ and $2C$
 (c) $2L$ and $C/2$
 (d) $2L$ and $2C$
- $\omega = \frac{1}{\sqrt{LC}}$
11. (3 points) Radio receivers are usually tuned by adjusting the capacitor of an LC circuit. If $C = C_1$ for a frequency of 600 kHz, then for a frequency of 1200 kHz one must adjust C to:
- (a) $C_1/2$
 (b) $C_1/4$
 (c) $2C_1$
 (d) $4C_1$
 (e) $\sqrt{2}C_1$
- $f = \frac{1}{2\pi\sqrt{LC}}$ $f \sim \frac{1}{\sqrt{C}}$ $C_0 = \frac{C_1}{4}$
12. (3 points) At least two transformers operate between us in Manhattan and the Jeffrey Energy Center in Bellevue, where our electric power originates. What kind of transformers are there?
- (a) Jeffrey: step-up. Manhattan: step-up.
 (b) Jeffrey: step-up. Manhattan: step-down.
 (c) Jeffrey: step-down. Manhattan: step-up
 (d) Jeffrey: step-down. Manhattan: step-down
13. (3 points) The rapid exponential decay in just a few cycles of the charge on the plates of capacitor in an RLC circuit might due to:
- (a) a large inductance
 (b) a large capacitance
 (c) a large resistance
 (d) a small capacitance
 (e) a small resistance

14. (3 points) The angular frequency of a certain RLC series circuit is ω_0 . A source of sinusoidal emf, with angular frequency ω , is inserted into the circuit and ω is varied while the amplitude of the source is held constant. For which of the following values of ω is the amplitude of the current oscillations the greatest?

(a) $\omega_0/5$
(b) $\omega_0/2$
(c) ω_0
(d) $2\omega_0$
(e) None of them (they all produce the same current amplitude)

15. (3 points) When the frequency of the oscillator in a series RLC circuit is doubled:

(a) the capacitive reactance is doubled —
(b) the capacitive reactance is halved ✓
(c) the impedance is doubled —
(d) the current amplitude is doubled —
(e) the current amplitude is halved —

$$X_C = \frac{1}{\omega C}$$

$$\omega' = 2\omega$$

$$X'_C = \frac{1}{2}X_C$$

$$X_L = \omega L$$

$$X'_L = 2X_L$$

$$X_R = R$$

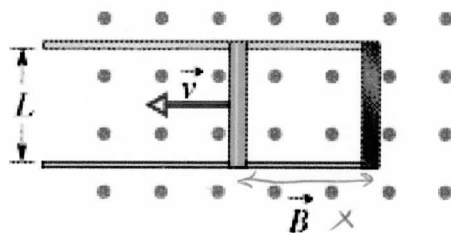
$$X'_R = X_R$$

16. (3 points) If one connects a 1.5 V "AA" battery to the primary coil of the transformer that has $N_1 = 100$ turns, what would be the potential difference in the secondary coil that has $N_2 = 1000$ turns?

(a) 1.5 V
(b) 15 V
(c) 0.15 V
(d) 0 V

Transformers only work with AC sources!

Part II: Work out this part in the space provided. Show your work!
(50 points total)



17. (12 points) In the figure, a metal rod is forced to move with constant velocity along two parallel metal rails, connected with a strip of metal at one end. A magnetic field of magnitude $B = 0.5 \text{ T}$ points out of the page.

- (a) (2 points) Assuming that at some point of time t the metal rod is at the distance x from the metal strip. Derive the magnetic flux through the loop formed by the metal rails, strip and the rod.

$$\Phi_B = BxL \quad (+2)$$

- (b) (3 points) If $L = 10 \text{ cm}$, and $v = 10 \text{ cm/s}$, what is the magnitude of the emf generated in volts?

$$\mathcal{E} = \left| \frac{d\Phi_B}{dt} \right| = BL \frac{dx}{dt} = BLv = 0.5 \text{ T} \cdot 0.1 \text{ m} \cdot 0.1 \frac{\text{m}}{\text{s}} = 5 \cdot 10^{-3} \text{ V} \quad (+1)$$

- (c) (3 points) If the rod has a resistance of 10Ω , and the rails and the strip have negligible resistance, what is the current in amperes in the rod?

$$I = \frac{\mathcal{E}}{R} = \frac{0.005 \text{ V}}{10 \Omega} = 5 \cdot 10^{-4} \text{ A} \quad (+2)$$

- (d) (4 points) What force is required to pull the metal rod to the right with a constant velocity? Neglect friction.

$$\text{Force on moving rod with current } I \text{ in } B: \quad F_B = ILB = 5 \cdot 10^{-4} \text{ A} \cdot 0.1 \text{ m} \cdot 0.5 \text{ T} = 2.5 \cdot 10^{-6} \text{ N} \quad (+2)$$

explanation: To have constant velocity, one needs to provide $F_{\text{pulling}} = F_B$ to have $F_{\text{NET}} = 0 \text{ N} \Rightarrow$

$$F_{\text{pulling}} = 2.5 \cdot 10^{-6} \text{ N} \quad (+1)$$

18. (8 points) A 10 cm long air-core solenoid consists of 20,000 turns of wire wrapped around a 1 cm^2 cross sectional area; 3 A of current pass through the coil.

(a) (2 points) Show that the magnetic field inside the coil has the value $B = 0.75 \text{ T}$.

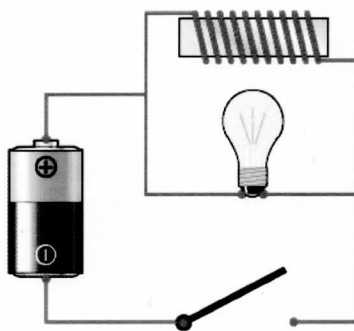
$$\textcircled{+1} \quad B = \mu_0 n I = 4\pi \cdot 10^{-7} \cdot \frac{20000}{0.1 \text{ m}} \cdot 3 \text{ A} = \textcircled{+1} 0.75 \text{ T}$$

(b) (3 points) Show that total flux through the coil, produced by the coil's own magnetic field is $\Phi_B = 1.5 \text{ T} \cdot \text{m}^2$.

$$\textcircled{+2} \quad \Phi_B = B \cdot A \cdot N = 0.75 \text{ T} \cdot 1 \text{ cm}^2 \cdot \frac{1 \text{ m}^2}{(100 \text{ cm})^2} \cdot 20000 = \textcircled{+1} 1.5 \text{ T} \cdot \text{m}^2$$

(c) (3 points) Determine the inductance of the coil.

$$\Phi = L I \quad \textcircled{+2}$$
$$L = \frac{\Phi}{I} = \frac{1.5 \text{ T} \cdot \text{m}^2}{3 \text{ A}} = \textcircled{+1} 0.5 \text{ H}$$



19. (12 points) In the circuit shown above, a 6 V battery with 5 Ω of internal resistance connects to the 0.5 H inductor with a negligible resistance and a 120 Ω resistor (the light bulb) through a switch.

(a) (3 points) Show that the light bulb draws 48 mA of current just after the switch is closed.

(+2) As soon as the switch is closed current flows through the bulb only

$$i = \frac{6V}{(120+5)\Omega} = 48 \cdot 10^{-3} A$$

(+1)

(b) (3 points) Show that the inductor draws 1.2 A of current a long time after the switch is closed.

(+2) { After the inductor is fully energized it offers no resistance to the current and all of it bypasses the light bulb and go through the inductor. So, effective resistance is 5 Ω $\Rightarrow$

(+1) $i = \frac{6V}{5\Omega} = 1.2 A$

(c) (3 points) Now suppose the switch is re-opened after being closed for a long time. Find the current in the light bulb just after this re-opening.

(+2) { Current would still be 1.2 A⁽⁺¹⁾ and it will flow through the light bulb

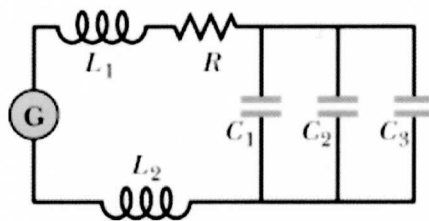
(d) (3 points) How long will it take for the current through the light bulb to drop to about a third of its maximum value after the switch is reopened?

$$\tau_{LR} = \frac{L}{R} = \frac{0.5 H}{120 \Omega} = 0.00423 \approx 4.2 \text{ ms.} \quad (+3)$$

Or: $I(t) = I_0 e^{-\frac{t}{L/R}} \Rightarrow \frac{I_0}{3} = I_0 e^{-\frac{t}{L/R}} \Rightarrow \frac{1}{3} = e^{-\frac{t}{L/R}} \Rightarrow$

(+2)

$$\frac{t}{L/R} = \ln 3 \Rightarrow t = \frac{L}{R} \ln 3 = 4.6 \text{ ms.} \quad (+1)$$



20. (18 points) In the circuit shown above, $R = 1.00 \Omega$, inductances $L_1 = 1.70 \text{ mH}$ and $L_2 = 2.30 \text{ mH}$, and capacitances $C_1 = 4.00 \mu\text{F}$, $C_2 = 2.50 \mu\text{F}$, and $C_3 = 3.50 \mu\text{F}$. The generator G provides an EMF of $V_0 \cos \omega_d t$, where $\omega_d = 2\pi f_d$. The peak voltage $V_0 = 171 \text{ V}$.

- (a) (2 points) What is the effective capacitance of the circuit?

$$C_{\text{eff}} = C_1 + C_2 + C_3 = 4 \mu\text{F} + 2.5 \mu\text{F} + 3.5 \mu\text{F} = 10 \mu\text{F}$$

(+1)

- (b) (3 points) Determine the capacitive reactance of the circuit for $f_d = 50 \text{ Hz}$.

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C} = \frac{1}{2\pi \cdot 50 \cdot 10 \cdot 10^{-6}} \Omega = 318.3 \Omega$$

(+2)

- (c) (2 points) What is the effective inductance of the circuit?

$$X_L = L_1 + L_2 = 1.7 \text{ mH} + 2.3 \text{ mH} = 4 \text{ mH}$$

(+1)

- (d) (3 points) Determine the inductive reactance of the circuit for $f_d = 50 \text{ Hz}$.

$$X_L = 2\pi f L = 2\pi \cdot 50 \cdot 4 \cdot 10^{-3} \Omega = 1.26 \Omega$$

- (e) (4 points) What is the resonant frequency of the circuit?

Using correct formula (+2)

$$\omega_0 = \frac{1}{\sqrt{LC}} = 5000 \frac{\text{rad}}{\text{s}} \text{ or } f = \frac{\omega_0}{2\pi} = 796 \text{ Hz}$$

correct units (+1)
answer (+1)

$$\text{or } \omega' = \sqrt{\omega_0^2 - \left(\frac{2R}{L}\right)^2} \approx 4975 \frac{\text{rad}}{\text{s}} \text{ or } f' = \frac{\omega'}{2\pi} \approx 792 \text{ Hz}$$

- (f) (4 points) What is the peak current in the circuit for the resonant frequency?

(+2) ← explanation is important!

$$I = \frac{\mathcal{E}}{R} \text{ as no contribution will be from } X_L \text{ and } X_C$$

(+1)

$$I = \frac{171 \text{ V}}{1 \Omega} = 171 \text{ A (wow!)}$$

Or

$$I = \frac{\mathcal{E}}{\sqrt{R^2 + (X_C - X_L)^2}} = \dots$$

should yield the same answer.
If student plugs values from (b) and (d) → (+1).